

Satwik Medipalli

AI Engineer — LLM Applications, RAG & Multi-Agent Systems

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SUMMARY

AI engineer with four live AI products built end to end during an M.S. at UT Dallas (2026) — multi-agent due diligence over SEC filings, statistically calibrated LLM-judge evaluation, and fleet-reliability ML with a claim-verifying LLM analyst — plus three internships shipping NLP pipelines and cloud ML deployments. Every project below is live at a public URL.

DEPLOYED PROJECTS (BUILT DURING M.S., 2024 – 2026)

Diligence AI: Multi-Agent Due Diligence Platform | *LangGraph, CrewAI, Pinecone* | diligence-ai-nine.vercel.app Spring 2026

- Built an autonomous due-diligence platform where six specialized agents read SEC 10-K filings, extract financials, classify risk factors, and assemble an analyst-style report in minutes
- Implemented a Pinecone-backed RAG interface so users can ask natural-language questions over analyzed filings and get source-cited answers, with RAGAS-scored faithfulness gating every release

VoiceIQ: Evaluation Platform for Voice AI Agents | *Python, LLM-as-judge, Streamlit* | [live site](https://live.site) Fall 2025

- Created a testing platform that simulates customer conversations across 10 scenario types using a two-LLM loop, scoring agent performance on five dimensions with weighted scoring and self-consistency tracking
- Validated the LLM judge itself with Bland-Altman agreement analysis, Pearson/Spearman correlation, and precision/recall tracking against reference scores, backed by 70 unit tests

SiliconSentinel: Fleet Reliability Intelligence | *Python, XGBoost, Pydantic AI* | silicon-sentinel.vercel.app Fall 2025

- Built a GPU fleet-reliability platform that catches hardware failures 30 days early: physics-grounded failure simulation (Arrhenius, Black, Peck, Coffin–Manson) across 10,000 devices at 12 sites feeds survival models and XGBoost remaining-useful-life scoring — 83% recall at 80% precision
- Shipped an ops console with an RMA-vs-monitor decision queue, SHAP deep dives, and a Pydantic AI analyst that verifies every numeric claim against fleet data before answering

NFL Draft Intelligence | *Python, XGBoost, SHAP, SQL* | nfl-draft-intelligence.vercel.app Spring 2025

- Trained position-specific XGBoost classifiers on two decades of NFL Combine data to evaluate draft prospects, with SHAP values explaining which measurables drive each prediction and a cosine-similarity engine matching prospects to historical comps

EXPERIENCE

AI Research Intern

Jan 2024 · Apr 2024

Boltzmann Labs

- Wrote a format-aware Python preprocessing engine that auto-detects three types of incoming biological data across ~500K records at 85%+ accuracy and routes each through the correct pipeline, replacing a fully manual data-preparation workflow
- Containerized the ML backend with Docker and deployed it on Kubernetes and AWS S3 behind REST APIs, turning a local prototype into a scalable cloud service for the frontend team

Data Analytics Intern

Aug 2023 · Dec 2023

Bostonlogix

- Built an NLP extraction pipeline with OpenAI APIs on Microsoft Azure that pulls structured fields from ~100K unstructured text records, fully automating a manual review process
- Migrated ~1M records from MongoDB to SQL Server with Python to create a queryable analytics store, and audited and restored roughly 1,000 failing API endpoints with Postman
- Designed Bold BI dashboards for Hungry Jack's that leadership used in shareholder reviews of store performance

BI Developer Intern

Apr 2023 · Jul 2023

Ideabytes

- Built a real-time ingestion pipeline that streamed ~500K JSON sensor datapoints through APIs into Power BI, and presented the live dashboard directly to the CEO
- Developed linear regression and ARIMA models forecasting sensor performance at 92–95% one-month accuracy and 80–90% at twelve months, letting the team schedule proactive maintenance before hardware failures actually occurred

EDUCATION

University of Texas at Dallas

Richardson, TX

M.S. Business Analytics

May 2026

GITAM University

Hyderabad, India

B.Tech Computer Science with Data Science

May 2024

TECHNICAL SKILLS

LLM & Agents: LangChain, LangGraph, CrewAI, OpenAI Agents SDK, Pydantic AI, RAG, Pinecone, RAGAS, LLM evaluation

ML: XGBoost, SHAP, Scikit-learn, survival analysis (lifelines), BERT/Transformers, forecasting, explainability

Engineering: Python, SQL, JavaScript, FastAPI, Flask, Docker, Kubernetes, AWS S3, Azure, Git, Postman

Data & Visualization: Pandas, NumPy, PostgreSQL, SQL Server, MongoDB, Power BI, Tableau, Streamlit, Plotly